

Milestone 3: Demand Intelligence & Mobility Analytics

WEEKS 5–6



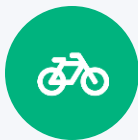
Objective: Understand mobility demand and identify changing travel patterns across ride-hailing, micromobility, and multimodal networks — turning raw trip data into forecasting-ready intelligence.

FOUR CORE COMPONENTS



Ride-Hailing

Trip demand, surge pricing, fares & peak-hour patterns



Micromobility

Bike-share utilization, turnover & rebalancing needs



Modal Substitution

How travelers switch between transport modes



Weather & Forecasting

Weather sensitivity and demand forecasting models



Ride-Hailing Intelligence

24.6K

Daily Trip Demand

▲ 12% vs last week

1.42x

Avg. Surge Multiplier

▲ 0.08x peak hours

₹238

Average Fare

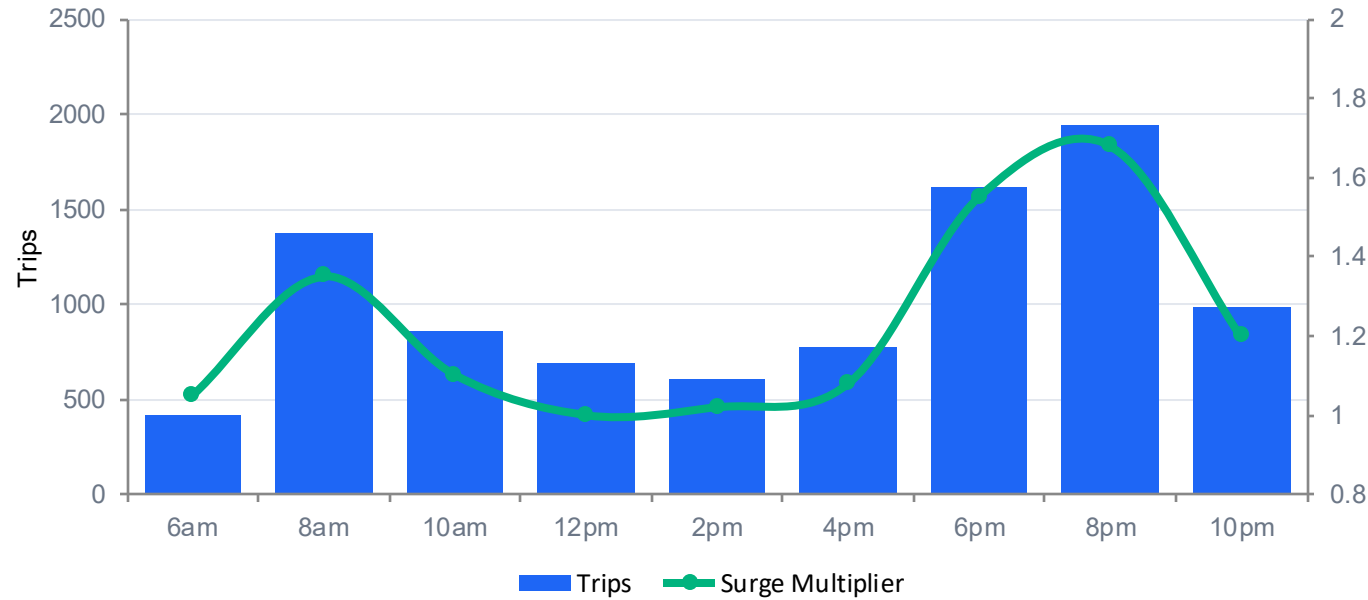
▲ 5% vs last week

6–9 PM

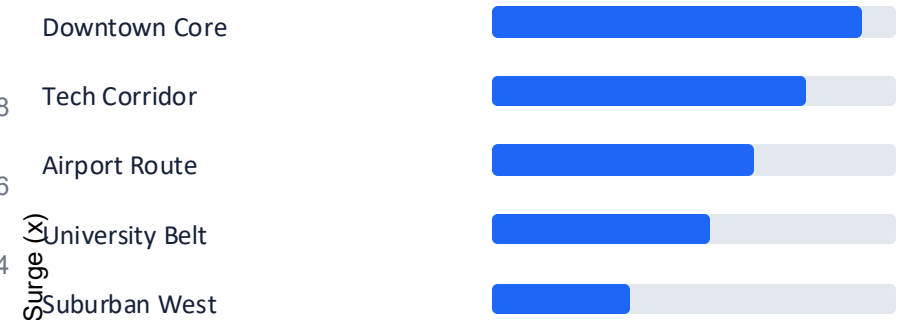
Peak-Hour Window

38% of daily demand

HOURLY TRIP DEMAND VS. SURGE MULTIPLIER



DEMAND BY ZONE



Demand Growth (WoW)

+18.4%

Sustained rise driven by evening commute + weekend leisure trips



Cycling & Micromobility Analytics

8,940

Bike-Share Trips Today

▲ 9% vs last week

3,210

Available Bikes (fleet)

72% fleet availability

68%

Avg. Station Utilization

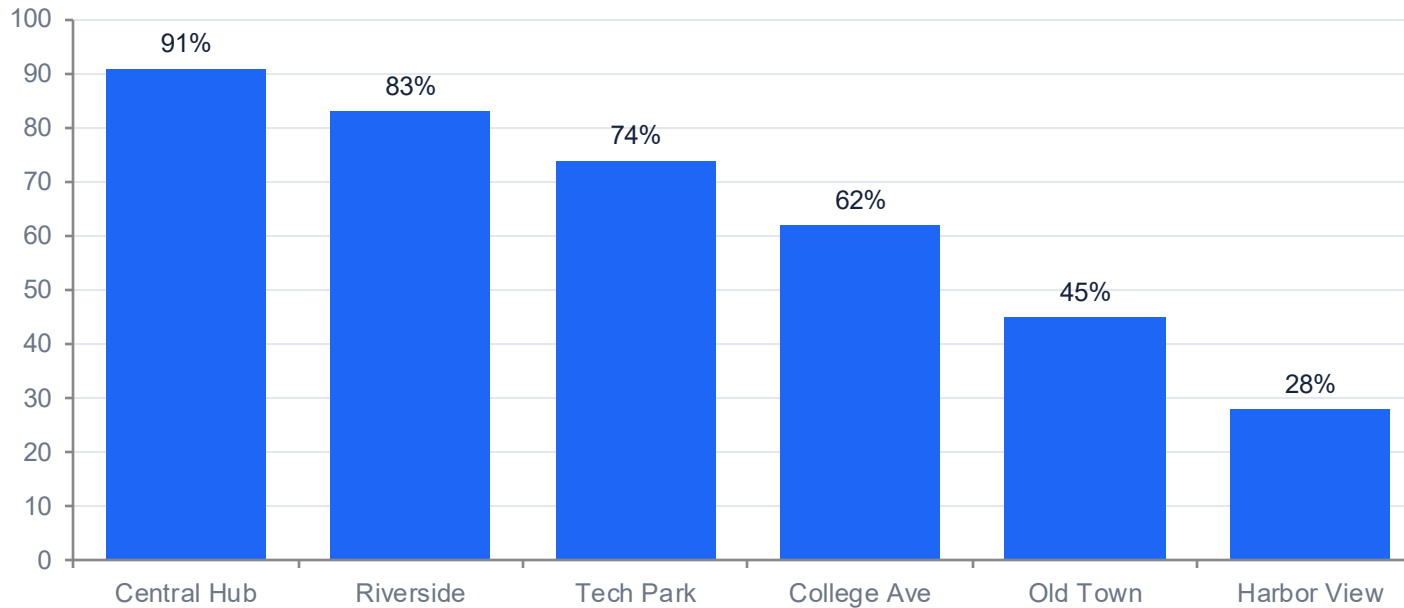
▲ 6 pts vs last week

4.6x

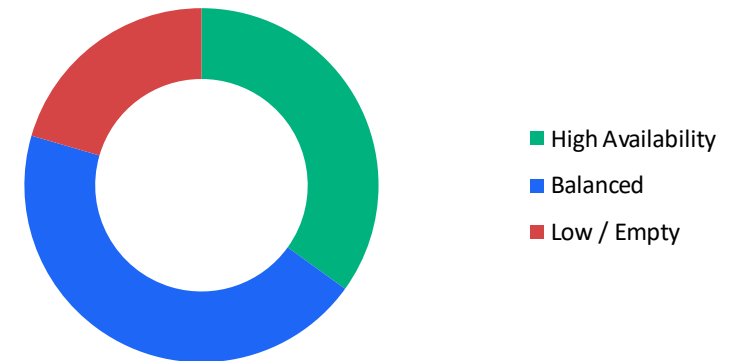
Daily Bike Turnover Rate

per dock, city-wide

STATION UTILIZATION BY ZONE



STATION AVAILABILITY MIX



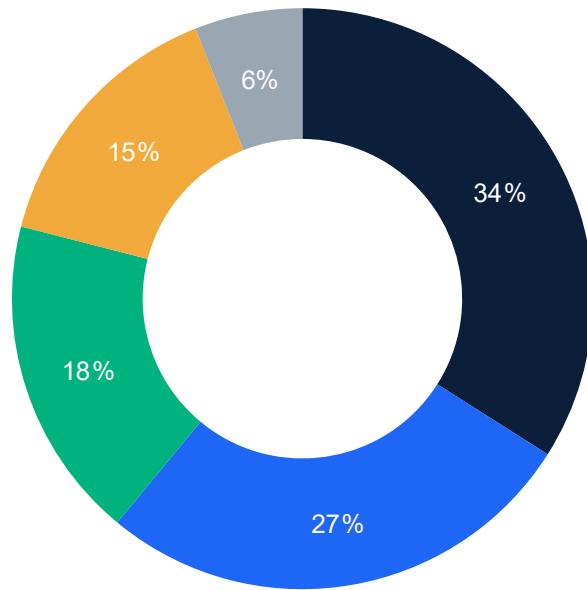
34 stations require rebalancing today — low-availability docks concentrated in Old Town & Harbor View.



Modal Substitution Analysis

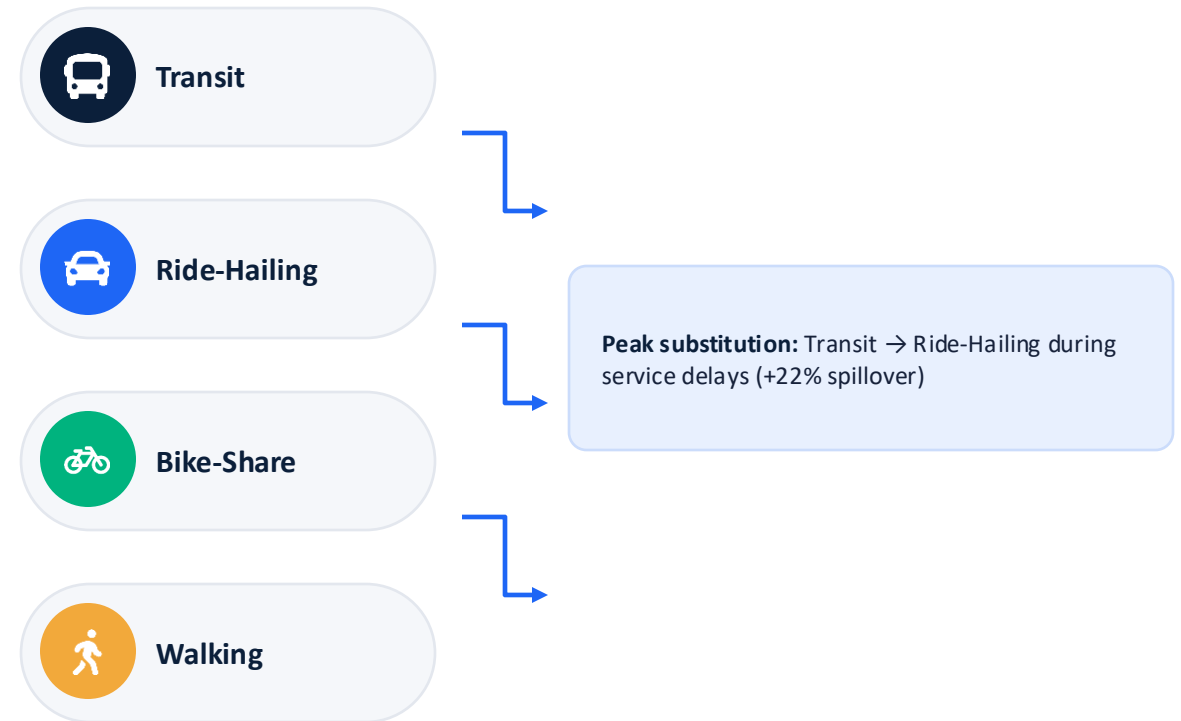
When ride-hailing surges, transit is delayed, or weather shifts, commuters substitute across modes. Tracking these switches reveals hidden demand elasticity across the network.

CURRENT MODAL SPLIT



■ Public Transit ■ Ride-Hailing ■ Bike-Share ■ Walking ■ Other Modes

MODAL SWITCHING FLOW





Weather Sensitivity & Demand Forecasting

94.2%

Forecast Model Accuracy

7-day rolling avg.

+22%

Weather-Driven Demand Swing

rain vs. clear-day baseline

+35%

Ride-Hailing Demand in Rain

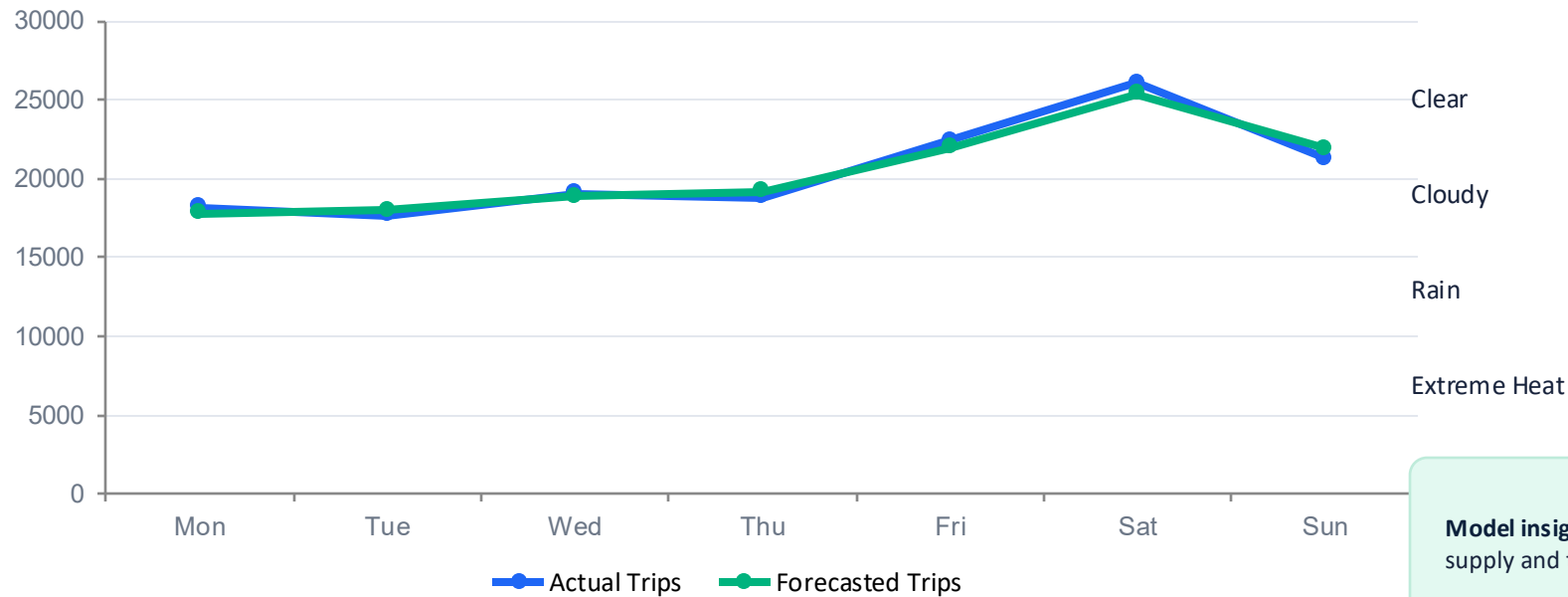
vs. clear conditions

-41%

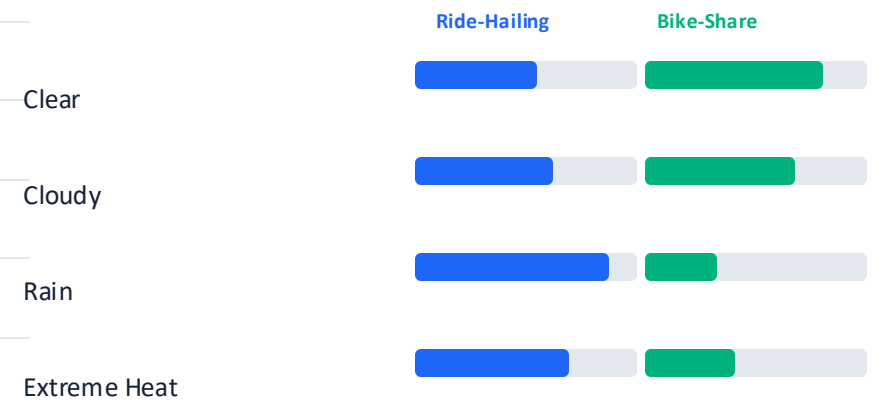
Bike-Share Demand in Rain

vs. clear conditions

7-DAY DEMAND FORECAST VS. ACTUAL TRIPS



DEMAND BY WEATHER CONDITION



Model insight: Rain forecasts 24h ahead can pre-position ride-hailing supply and trigger bike-share rebalancing.